

2.2.win32), Matplot Library (matplotlib-0.91.2.win32-Py2.5) and Numeric Python (numpy-1.0.4.win32-py2.5). Output pattern is displayed using Tkinter software, and values are displayed in a text file. Tkinter software is a part of Python package, so there is no need to separately install it.

Testing procedure

To test the project, follow the steps given below:

Step 1. Install all required software including Python2.5, Win32 Python, pyserial-2.2, matplotlib-0.91 and numpy-1.0.4 on your Windows 7 operating system.

Step 2. Power on the board (GND, 5V, 12V) and connect it to the computer through serial COM port. If physical 9-pin serial COM port is not available, use a serial-to-USB converter.

Step 3. The physical or enumerated COM port number that appears in Device Manager should be written in max5154.py script (self.fd = serial.Se-

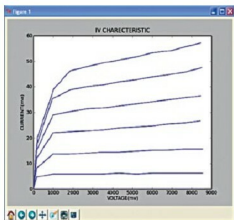


Fig. 7: I-V characteristic curve of BC107 transistor

rial('\\\\.\\COM18', 115200, stopbits = 1, parity = serial.PARITY_NONE))

Step 4. Place efpython and max5154 Python programs in python25 folder (under C drive), where the microcontroller is already programmed.

Step 5. Open PythonWin IDE.

Step 6. Open efpython program from pythonWin IDE. It should be in Python25 folder. Run the program using Run from Menu bar.

Step 7. If Python script is run-

EFY Note

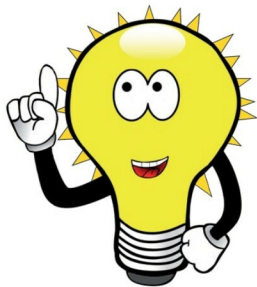
The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

ning, wait for a minute. The program will start showing the current and voltage values on the screen. Once measurement is over, the graph is displayed automatically. You will get the output screen as shown in Fig. 7.

The data acquisition system using Python interface can be used as a starting point for interfacing any gadget with suitable modifications. The interactive nature of Python software makes it more user friendly as compared to other visual languages. **EFY**



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